A PROPER BIRATIONAL MORPHISM OF 2-DIMENSIONAL NAGATA SPACES IS DOMINATED BY A SEQUENCE OF NORMALIZED BLOWUPS IN CLOSED POINTS

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Editorial note. Two same-chapter inputs are NOT carded and are declared external: lemma-equivalence-sequence-normalized-blowups (zero-content cross-reference to the blowup-sequence proof) and lemma-equivalence (dropped: fully-faithfulness part explicitly omitted).

Allowed standard prerequisites (editorial).

- Normalizations are normal and morphisms factor through them. Morphisms of Spaces, section normalization (cross-chapter, named). Lets the proof replace X and Y by their normalizations.
- Modifications of normal spaces are isomorphisms off a finite set of closed points. Spaces over Fields, lemma modification-normal-iso-over-codimension-1 (cross-chapter, named). Produces the finite set T with f an isomorphism over X minus T.
- Dominating modifications by admissible blowups. More on Morphisms of Spaces, lemma dominate-modification-by-blowup (cross-chapter, named). Gives an X minus T admissible blowup Y' -> X dominating Y, made representable after normalization.
- Scheme-case domination by normalized blowups. Resolution of Surfaces, lemma dominate-by-normalized-blowing-up (cross-chapter, named). Provides the normalized blowup sequence on each etale neighbourhood dominating Y_i.
- Normalized blowup sequences under the etale neighbourhood equivalence. lemma-equivalence-sequence-normalized-blowups (same chapter, NOT carded, declared external). Assembles the per-neighbourhood sequences into one global sequence.
- The etale neighbourhood functor is an equivalence. lemma-equivalence (same chapter, NOT carded, declared external). Concludes X_m dominates Y from the per-neighbourhood domination.

Source reference. Original reference 0BBP: focused-seeds/geometry/S9-spres-dominate-by-normalized-blowing-up/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BHB: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 425–425. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original lemma 0BHL: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 708–708. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BHM: focused-seeds/geometry/S9-spres-dominate-by-normalized-blowing-up/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 0BH9: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 391–391. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 087G: focused-seeds/geometry/S9-spres-dominate-by-normalized-blowing-up/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAW: focused-seeds/geometry/S9-spres-dominate-by-normalized-blowing-up/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAT: focused-seeds/geometry/S9-spres-dominate-by-normalized-blowing-up/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07U3: focused-seeds/geometry/S9-spres-dominate-by-normalized-blowing-up/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BBQ: focused-seeds/geometry/S9-spres-dominate-by-normalized-blowing-up/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian Nagata algebraic space over S with $\dim(X) = 2$. Let $f : Y \to X$ be a proper birational morphism. Then*

there exists a commutative diagram

$$\begin{array}{ccccccc} X_n & \longrightarrow & X_{n-1} & \longrightarrow & \ldots & \longrightarrow & X_1 & \longrightarrow & X_0 \\ \downarrow & & & & & & & & \downarrow \\ Y & & & & & & & & X \end{array}$$

where $X_0 \to X$ is the normalization and where $X_{i+1} \to X_i$ is the normalized blowing up of X_i at a closed point.

Proof. Although one can prove this lemma directly for algebraic spaces, we will continue the approach used above to reduce it to the case of schemes.

We will use that Noetherian algebraic spaces are quasi-separated and hence points have well defined residue fields (for example by Decent Spaces, Lemma 0BBP). We will use the results of Morphisms of Spaces, Sections 0BAT, 0BAW, and 07U3 without further mention. We may replace Y by its normalization. Let $X_0 \to X$ be the normalization. The morphism $Y \to X$ factors through X_0. Thus we may assume that both X and Y are normal.

Assume X and Y are normal. The morphism $f : Y \to X$ is an isomorphism over an open which contains every point of codimension 0 and 1 in Y and every point of Y over which the fibre is finite, see Spaces over Fields, Lemma 0BBQ. Hence we see that there is a finite set of closed points $T \subset |X|$ such that f is an isomorphism over $X \setminus T$. By More on Morphisms of Spaces, Lemma 087G there exists an $X \setminus T$-admissible blowup $Y' \to X$ which dominates Y. After replacing Y by the normalization of Y' we see that we may assume that $Y \to X$ is representable.

Say $T = \{x_1, \ldots, x_r\}$. Pick elementary étale neighbourhoods $(U_i, u_i) \to (X, x_i)$ as in Section 0BH9. For each i the morphism $Y_i = Y \times_X U_i \to U_i$ is a proper birational morphism which is an isomorphism over $U_i \setminus \{u_i\}$. Thus we may apply Resolution of Surfaces, Lemma 0BHM to find a sequence

$$X_{i,m_i} \to X_{i,m_i-1} \to \ldots \to X_1 \to X_{i,0} = U_i$$

of normalized blowing ups in closed points lying over u_i such that X_{i,m_i} dominates Y_i. By Lemma 0BHL we find a sequence of normalized blowing ups

$$X_m \to X_{m-1} \to \ldots \to X_1 \to X_0 = X$$

as in the statement of the lemma whose base change to our U_i produces the given sequences. It follows that X_m dominates Y by the equivalence of categories of Lemma 0BHB. □